Luxury Housing Market Analysis Dashboard

Project Documentation

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Course: Data Science – GUVI

Project Type: End-to-End Data Analytics Project

1. Project Overview

This project focuses on analysing luxury housing market data to identify patterns in bookings, pricing, builder performance, and geographical demand.

The main objective of this project is to transform raw housing data into meaningful insights using Python, SQL, and Power BI.

Python was used for data cleaning and preparation. SQL was used to create tables and validate data through queries. Power BI was used to create interactive dashboards that help visualize business trends.

The dashboard provides insights into booking conversion trends, configuration demand, sales channel performance, builder contributions, and geographical market analysis.

2. Project Pipeline Architecture

The project follows a structured data analytics pipeline to transform raw data into meaningful insights.

Step 1: Data Collection

Raw housing data was collected in CSV format containing information about builders, pricing, configurations, sales channels, and geographical locations.

Step 2: Data Cleaning using Python

Python was used to clean the dataset by removing null values, correcting data types, and standardizing column names.

Step 3: Feature Creation

Additional features were created to improve analysis, such as booking indicators and formatted quarter values.

Step 4: Load Data into SQL Database

The cleaned dataset was loaded into a SQL database to enable structured querying and validation.

Step 5: Query Data using SQL

SQL queries were executed to validate data, count records, group booking status, and calculate average ticket prices.

Step 6: Create Dashboard using Power BI

Power BI was used to create interactive dashboards to visualize booking trends, revenue, builder performance, and geographical insights.

Step 7: Generate Business Insights

Final insights were generated from visual dashboards to support data-driven decision-making.

3. Python Data Processing

Python was used for cleaning and preparing the dataset before loading it into the SQL database.

The following Python scripts were created and used in this project:

data_cleaning.py
This script was used to clean the dataset by removing missing values, correcting data types, and standardizing column names. It ensured that the dataset was ready for analysis.

mysql_import.py
This script was used to connect Python to the MySQL database and import the cleaned dataset into the SQL table.

Key tasks performed using Python:

- Loaded dataset from CSV file
- Handled missing values
- Converted column data types
- Saved cleaned dataset as cleaned_luxury_housing.csv
- Connected Python to MySQL

- Imported cleaned dataset into SQL database

```python

import pandas as pd

# Step 1: Read the dataset
df = pd.read_csv("Luxury_Housing_Bangalore.csv")

# Step 2: Display first 5 rows
print(df.head())

# Step 3: Show column names
print("Columns in dataset:")
print(df.columns)

# Step 4: Show dataset info
print(df.info())

# Step 5: Check missing values
print("\nMissing values in each column:")
print(df.isnull().sum())

# Step 6: Handle missing values

# Fill Amenity_Score with median
df["Amenity_Score"] = df["Amenity_Score"].fillna(
    df["Amenity_Score"].median()
```

```python
# Fill Buyer_Comments with text
df["Buyer_Comments"] = df["Buyer_Comments"].fillna(
    "No Comments"
)

print("\nMissing values after cleaning:")
print(df.isnull().sum())

# Step 7: Standardize text columns

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text_columns = [
    "Transaction_Type",
    "Buyer_Type",
    "Purchase_Quarter",
    "Possession_Status",
    "Sales_Channel",
    "NRI_Buyer",
    "Buyer_Comments"
]

for col in text_columns:
    df[col] = df[col].astype(str).str.strip().str.title()

print("\nText columns standardized successfully.")
```

```python
# Step 8: Create new feature columns

# Create Quarter_Number column
# Step 8: Create Quarter_Number from date

# Convert Purchase_Quarter to datetime
df["Purchase_Quarter"] = pd.to_datetime(df["Purchase_Quarter"])

# Extract quarter number
df["Quarter_Number"] = df["Purchase_Quarter"].dt.quarter

print("\nQuarter_Number column created successfully.")

# Check output
print(df[["Purchase_Quarter", "Quarter_Number"]].head())

# Step 9: Save cleaned dataset

df.to_csv("cleaned_luxury_housing.csv", index=False)

print("\nCleaned dataset saved successfully as 'cleaned_luxury_housing.csv'")

print("\nFinal columns:")
print(df.columns)

print("\nFirst 5 rows of cleaned data:")
print(df.head())
```

4. SQL Database Implementation

SQL was used to create structured tables and validate the dataset after loading it from Python.

The following SQL files were created and used in this project:

table_schema.sql

This file contains the SQL query used to create the housing data table. The table stores all cleaned housing dataset fields such as developer name, micro-market, ticket price, possession status, and booking information.

insert_queries.sql

This file contains SQL insert commands used to insert data into the housing data table.

analysis_queries.sql

This file contains SQL validation queries used to check the dataset and perform aggregation operations.

<u>Key SQL queries executed:</u>

- SELECT COUNT(*) — Used to count total number of records
- GROUP BY Possession_Status — Used to analyze booking distribution
- AVG(Ticket_Price_Cr) — Used to calculate average ticket price per builder

SQL ensured structured data storage and accurate validation before visualization.

```
CREATE DATABASE luxury_housing_db;
USE luxury_housing_db;
```

```
);

SELECT COUNT(*) FROM luxury_housing_data;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Con

COUNT(*)
101000

```
CREATE TABLE luxury_housing_data (

    Property_ID VARCHAR(50),
    Micro_Market VARCHAR(100),
    Project_Name VARCHAR(100),
    Unit_Size_Sqft FLOAT,
    Ticket_Price_Cr FLOAT,
    Configuration VARCHAR(50),
    Purchase_Quarter DATE,
    Buyer_Type VARCHAR(50),
    Transaction_Type VARCHAR(50),
    Connectivity_Score FLOAT,
    Amenity_Score FLOAT,
    Possession_Status VARCHAR(50),
    Sales_Channel VARCHAR(50),
    NRI_Buyer VARCHAR(20),
    Locality_Infra_Score FLOAT,
    Avg_Traffic_Time_Min INT,
    Buyer_Comments TEXT,
    Quarter_Number INT

);
```

```
SELECT
    Possession_Status,
    COUNT(*) AS total_properties
FROM luxury_housing_data
GROUP BY Possession_Status
ORDER BY total_properties DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Conte

Possession_Status	total_properties
Under Construction	33769
Launch	33641
Ready To Move	33590

```
SELECT
    Developer_Name,
    ROUND(AVG(Ticket_Price_Cr),2) AS avg_ticket_price
FROM luxury_housing_data
GROUP BY Developer_Name
ORDER BY avg_ticket_price DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Developer_Name	avg_ticket_price
Sobha	10.34
Godrej	10.28
Prestige	10.26
Total Environment	10.22
L&T Realty	10.22
Puravankara	10.19
RMZ	10.14
Tata Housing	10.13
Brigade	10.08
SNN Raj	10.07
Embassy	10.01

5. Power BI Dashboard Development

Power BI was used to create interactive dashboards that visualize the housing dataset and generate meaningful insights.

The cleaned data stored in the SQL database was connected to Power BI for visualization.

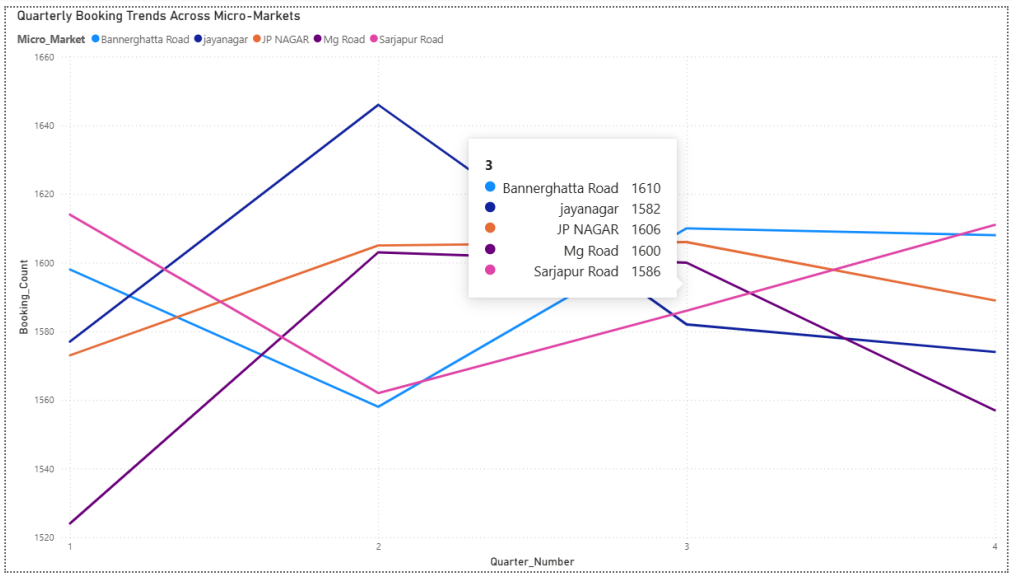
Multiple visualizations were created to analyse different aspects of the housing market.

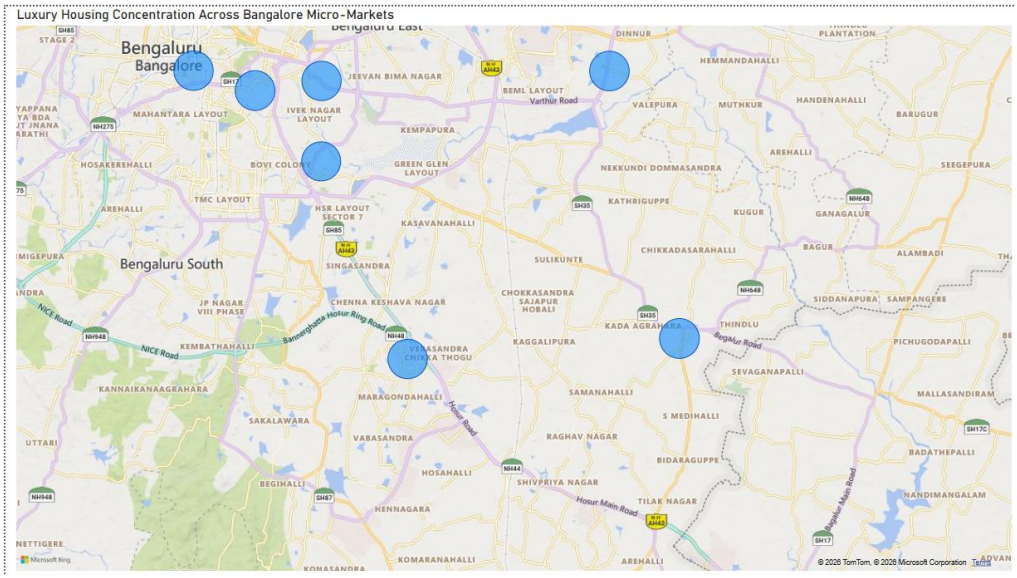
The following dashboards were developed in this project:

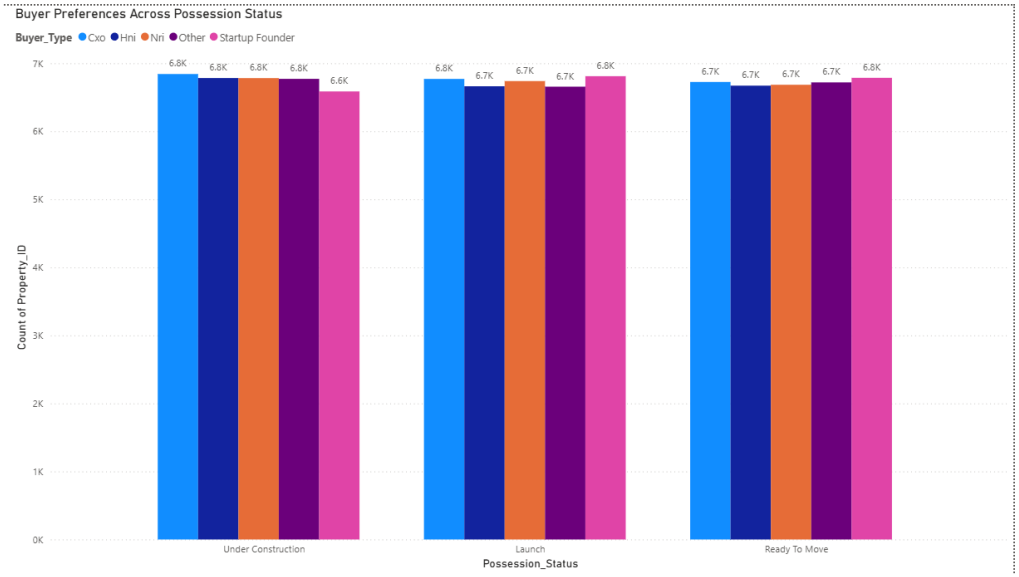
- Booking Conversion Analysis
- Configuration Demand Analysis
- Sales Channel Efficiency
- Quarterly Builder Contribution
- Possession Status Analysis
- Geographical Insights
- Top Performers Dashboard

Key features of the Power BI dashboard:

- Interactive filters and slicers
- KPI cards showing total revenue and bookings
- Bar charts and column charts for comparison
- Map visualization for geographical insights
- Matrix visualization for quarterly performance
- Top performers analysis using filters







6. Business Insights and Observations

The following key business insights were generated from the Power BI dashboard analysis:

1. High Demand for Mid-Sized Configurations

The analysis shows that 3BHK configurations have the highest number of bookings compared to other configurations. This indicates that buyers

prefer mid-sized housing units that offer a balance between affordability and space.

Real-world impact:
Developers can focus more on constructing 3BHK units to match buyer demand and improve sales conversion.

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2. Top Builders Dominate Market Revenue
Builders such as Prestige, L&T Realty, Total Environment, Godrej, and SNN Raj generated the highest total revenue among all developers.

Real-world impact:
These builders likely have strong brand trust and quality projects, making them market leaders. Smaller builders can study their strategies to improve competitiveness.

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3. Certain Micro-Markets Show Higher Demand
Some Bangalore micro-markets show higher project concentration and booking levels, as seen in the geographical map visualization.

Real-world impact:
Developers and investors can focus on high-demand micro-markets to maximize returns and reduce project risk.

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4. Sales Channel Efficiency Varies
Analysis of sales channels shows that certain channels contribute more bookings than others, indicating higher efficiency.

Real-world impact:

Real estate companies can allocate more marketing budget to high-performing sales channels and reduce spending on low-performing channels.

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5. Buyers Prefer Ready-to-Move Properties

Possession status analysis indicates higher booking counts for ready-to-move properties compared to under-construction projects.

Real-world impact:
Developers should maintain faster project completion timelines to attract more buyers.

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6. Builder Performance is Consistent Across Quarters

Quarterly contribution analysis shows that certain developers maintain consistent revenue performance across multiple quarters.

Real-world impact:
Consistent performance indicates financial stability and strong customer demand for these builders.

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7. Revenue Distribution Helps Identify Market Leaders

Top performers visualization clearly identifies builders generating the highest revenue.

Real-world impact:
Investors can use this information to partner with high-performing builders for future projects.

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8. Geographical Clustering Indicates Investment Hotspots

The map visualization shows clusters of projects in specific micro-markets.

Real-world impact:
These clusters indicate infrastructure development and strong demand, making them ideal locations for future investment.

7. Challenges Faced and Solutions

During the development of this project, several challenges were encountered. These challenges were addressed using proper debugging and systematic problem-solving.

1. Handling Missing Values in Dataset

Some columns contained missing or inconsistent values which affected data accuracy.

Solution:
Python was used to clean the dataset by removing or correcting missing values and standardizing column formats.

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2. Converting Data Types Correctly

Certain columns had incorrect data types which caused errors during SQL loading and visualization.

Solution:
Python scripts were used to convert columns into correct numeric and categorical formats before importing into SQL.

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3. Creating Structured SQL Tables

Designing the SQL table structure required careful mapping of dataset columns to correct data types.

Solution:
A structured table schema was created using SQL, and validation queries were executed to ensure data integrity.

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4. Selecting Appropriate Visualizations
Choosing the correct chart type for each analysis was initially challenging.

Solution:
Different visualizations were tested in Power BI, and the most meaningful charts were selected based on clarity and data representation.

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5. Map Visualization Configuration
Initially, map visuals did not load correctly due to location settings and disabled map permissions.

Solution:
Map permissions were enabled in Power BI settings, and Micro_Market was categorized as a city to correctly display locations.

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6. Organizing Dashboard Layout
Arranging multiple visuals in a readable and professional format required careful planning.

Solution:
Visuals were arranged logically across different pages with proper titles and formatting to improve readability.

8. Project Conclusion

This project successfully demonstrated the complete data analytics pipeline from raw data processing to business insight generation.

Using Python, the dataset was cleaned and prepared for analysis. SQL was used to store structured data and validate records using queries. Power BI was used to create interactive dashboards that helped visualize trends and patterns within the luxury housing market.

The dashboard provides valuable insights into builder performance, booking trends, configuration demand, and geographical market behaviour. These insights can help real estate developers, investors, and decision-makers identify high-performing builders, understand buyer preferences, and focus on high-demand locations.

Overall, this project highlights the importance of data-driven decision-making in the real estate industry and demonstrates practical skills in Python, SQL, and Power BI.